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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 18th March Morning Shift

JEE Main Question Paper with Answer Key

18 Mar 2021	90	17	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	22 questions
Physics	27 questions
Out of syllabus	17 questions
Answer key	at the end

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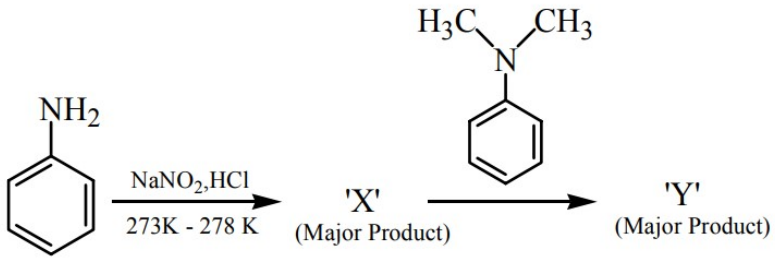
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Chemistry 24 questions

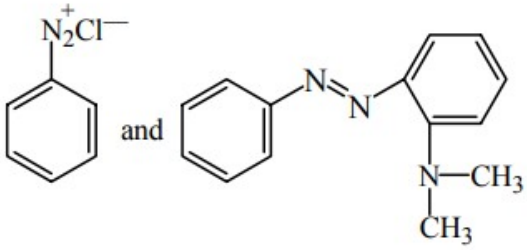
Q1. MCQ (Single Correct Answer)

Open solution

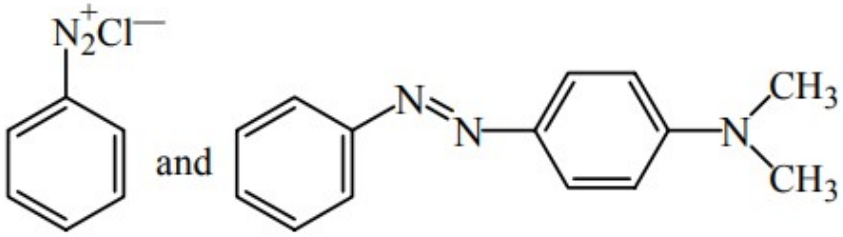


Considering the above reaction, X and Y respectively are :

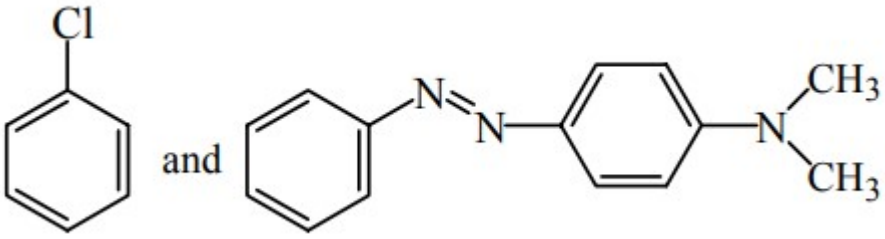
(A)



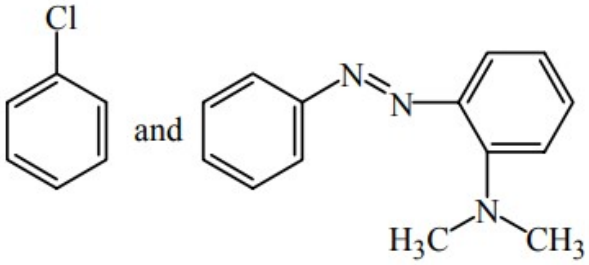
(B)



(C)



(D)



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Q2. MCQ (Single Correct Answer)

The ionic radius of Na^+ ions is 1.02 $\overset{o}{A}$. The ionic radii (in $\overset{o}{A}$) of Mg^{2+} and Al^{3+}, respectively, are

(A) 1.05 and 0.99 (B) 0.72 and 0.54 (C) 0.85 and 0.99 (D) 0.68 and 0.72

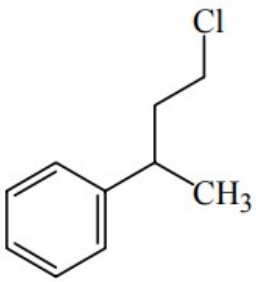
Open solution

Q3. MCQ (Single Correct Answer)

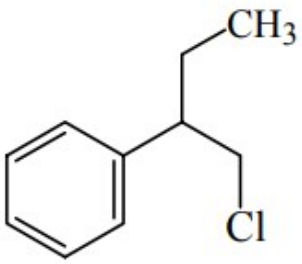
Reaction of Grignard reagent, C_2H_5MgBr with C_8H_8O followed by hydrolysis gives compound "A" which reacts instantly with Lucas reagent to give compound B, $C_{10}H_{13}Cl$. The Compound B is :

Open solution

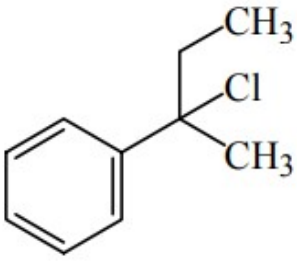
(A)



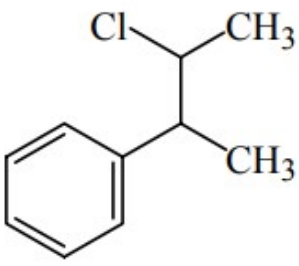
(B)



(C)



(D)



Q4. MCQ (Single Correct Answer)

Reagent, 1-napthylamine and sulphanilic acid in acetic acid is used for the detection of

(A) N_2O (B) NO_3^- (C) NO (D) NO_2^-

Open solution

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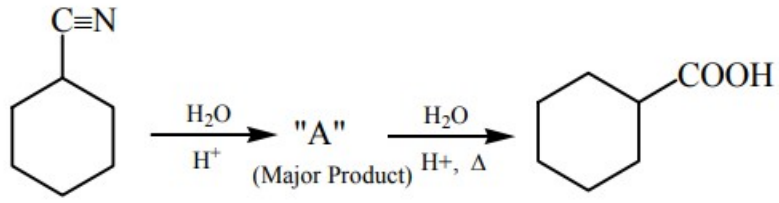
Q5. MCQ (Single Correct Answer)

A non-reducing sugar "A" hydrolyses to give two reducing mono saccharides. Sugar A is

(A) Fructose (B) Galactose (C) Glucose (D) Sucrose

Open solution

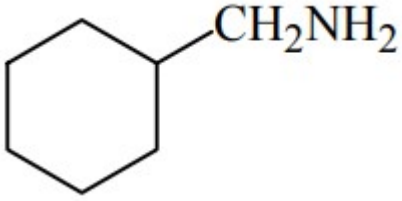
Q6. MCQ (Single Correct Answer)



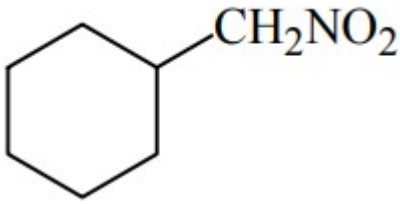
Open solution

Consider the above chemical reaction and identify product "A"

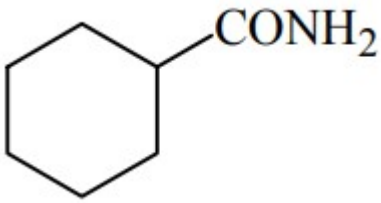
(A)



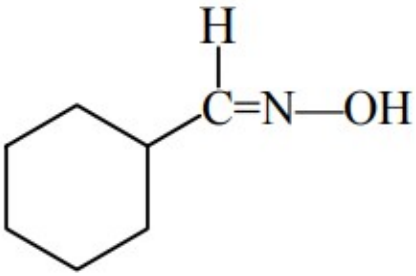
(B)



(C)



(D)



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Q7. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II

	List - I		List - II
(a)	Chlorophyll	(i)	Ruthenium
(b)	Vitamin-B_{12}	(ii)	Platinum
(c)	Anticancer drug	(iii)	Cobalt
(d)	Grubbs catalyst	(iv)	Magnesium

Choose the most appropriate answer from the options given below :

(A) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

(B) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

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Q8. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

List - I (Chemicals)

(a) Alcoholic potassium hydroxide

(b) $Pd/BaSO_4$

(c) BHC (Benzene hexachloride)

(d) Polyacetylene

List - II (Use / Preparation / Constituent)

(i) Electrodes in batteries

(ii) Obtained by addition reaction

(iii) Used for β - elimination reaction

(iv) Lindlar's catalyst

Choose the most appropriate match :

(A) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)
(B) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
(C) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)
(D) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Q9. MCQ (Single Correct Answer)

Open solution

Compound with molecular formula C_3H_6O can show :

(A) Positional isomerism
(B) Both positional isomerism and metamerism
(C) Metamerism
(D) Functional group isomerism

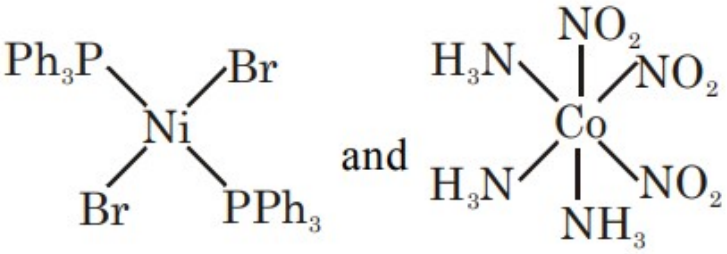
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Q10. MCQ (Single Correct Answer)

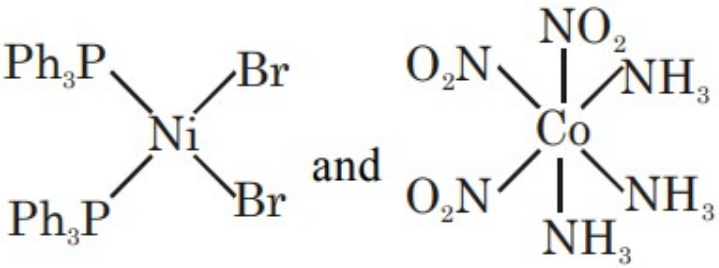
The correct structures of trans-$[NiBr_2(PPh_3)_2]$ and meridional-$[Co(NH_3)_3(NO_2)_3]$, respectively, are

Open solution

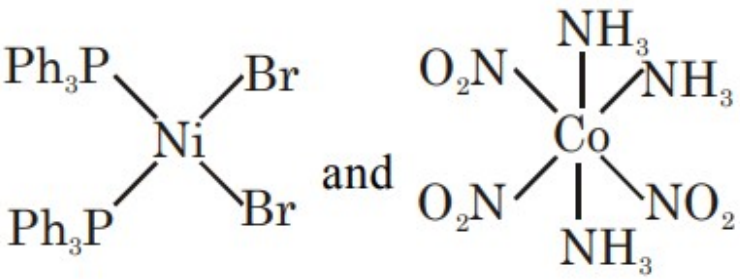
(A)



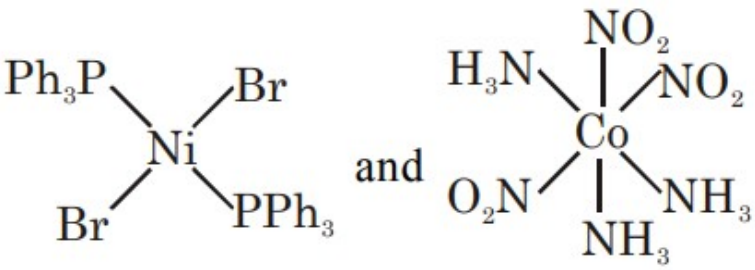
(B)



(C)



(D)



Q11. MCQ (Single Correct Answer)

A certain orbital has no angular nodes and two radial nodes. The orbital is :

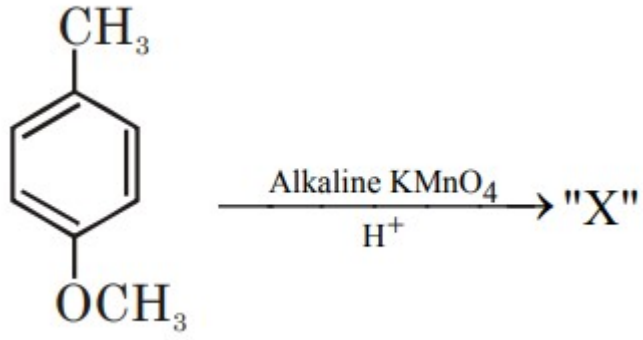
(A) 2s (B) 3s (C) 3p (D) 2p

Open solution

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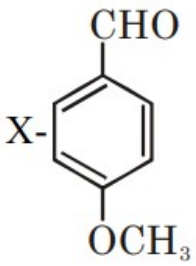
Q12. MCQ (Single Correct Answer)

Open solution

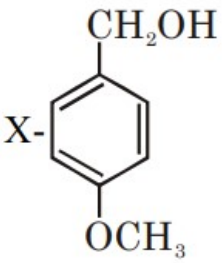


Considering the above chemical reaction, identify the product "X" :

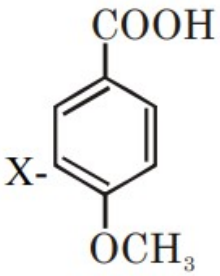
(A)



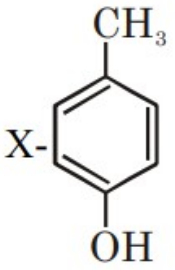
(B)



(C)



(D)



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Q13. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II

	List - I (process)		List - II (catalyst)
(a)	Deacon's process	(i)	ZSM-5
(b)	Contact process	(ii)	$CuCl_2$
(c)	Cracking of hydrocarbons	(iii)	Particles 'Ni'
(d)	Hydrogenation of vegetable oils	(iv)	V_2O_5

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(B) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(C) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(D) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q14. MCQ (Single Correct Answer)

Open solution

Given below are two statements : One is labelled as Assertion A and the other labelled as reason R

Assertion A : During the boiling of water having temporary hardness, $Mg(HCO_3)_2$ is converted to $MgCO_3$.

Reason R : The solubility product of $Mg(OH)_2$ is greater than that of $MgCO_3$.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are true but R is not the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

Q15. Numerical

Open solution

AX is a covalent diatomic molecule where A and X are second row elements of periodic table. Based on Molecular orbital theory, the bond order of AX is 2.5. The total number of electrons in AX is _____. (Round off to the Nearest Integer).

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Q16. Numerical

In order to prepare a buffer solution of pH 5.74, sodium acetate is added to acetic acid. If the concentration of acetic acid in the buffer is 1.0 M, the concentration of sodium acetate in the buffer is _____ M. (Round off to the Nearest Integer). [Given : pKa (acetic acid) = 4.74]

Open solution

Q17. Numerical

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(s)$

This reaction was studied at -10° and the following data was obtained

Run	$[NO]_0$	$[Cl_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall order of the reaction is _____. (Round off to the Nearest Integer).

Open solution

Q18. Numerical

For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$

the reaction enthalpy $\Delta_r H$ = _____ kJ mol^{-1}. (Round off to the Nearest Integer).

[Given : Bond enthalpies in kJ mol^{-1} : C-C : 347, C = C : 611; C-H : 414, H-H : 436]

Open solution

Q19. Numerical

_____ grams of 3-Hydroxy propanal (MW = 74) must be dehydrated to produce 7.8 g of acrolein (MW = 56) (C_3H_4O) if the percentage yield is 64. (Round off to the Nearest Integer).

[Given : Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

Open solution

Q20. Numerical

A reaction of 0.1 mole of Benzylamine with bromomethane gave 23 g of Benzyl trimethyl ammonium bromide. The number of moles of bromomethane consumed in this reaction are n $\times$ 10^{-1}, when n = _____. (Round off to the Nearest Integer).

[Given : Atomic masses : C : 12.0 u, H : 1.0 u, N : 14.0 u, Br : 80.0 u]

Open solution

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Q21. Numerical

The total number of unpaired electrons present in the complex $K_3[Cr(oxalate)_3]$ is _____.

Open solution

Q22. Numerical

2 molal solution of a weak acid HA has a freezing point of -3.885°C. The degree of dissociation of this acid is _____ $\times 10^{-3}$. (Round off to the Nearest Integer).

[Given : Molal depression constant of water = 1.85 K kg mol^{-1}, Freezing point of pure water = 0° C]

Open solution

Q23. Numerical

For the reaction

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)$

the magnitude of the standard molar Gibbs free energy change, $\Delta_r G^o_m$ = – _____ kJ (Round off to the Nearest Integer).

$$\left[\begin{matrix} E^o_{Fe^{2+}/Fe(s)} = -0.440V; & E^o_{Fe^{3+}/Fe(s)} = -0.036V \\ E^o_{I_2/2I^-} = 0.539V; & F = 96500C \end{matrix}\right]$$

Open solution

Q24. Numerical

Complete combustion of 3g of ethane gives x $\times 10^{22}$ molecules of water. The value of x is _____. (Round off to the Nearest Integer). [Use : N_A = 6.023 $\times 10^{23}$; Atomic masses in u : C : 12.0; O : 16.0; H : 1.0]

Open solution

Mathematics 22 questions

Q25. MCQ (Single Correct Answer)

The number of integral values of m so that the abscissa of point of intersection of lines 3x + 4y = 9 and y = mx + 1 is also an integer, is :

(A) 1 (B) 2 (C) 3 (D) 0

Open solution

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Q26. MCQ (Single Correct Answer)

The solutions of the equation $\begin{vmatrix} 1+\sin^2 x & \sin^2 x & \sin^2 x \\ \cos^2 x & 1+\cos^2 x & \cos^2 x \\ 4\sin 2x & 4\sin 2x & 1+4\sin 2x \end{vmatrix} = 0, (0 < x < \pi)$, are

Open solution

(A) $\frac{\pi}{12}, \frac{\pi}{6}$

(B) $\frac{\pi}{6}, \frac{5\pi}{6}$

(C) $\frac{5\pi}{12}, \frac{7\pi}{12}$

(D) $\frac{7\pi}{12}, \frac{11\pi}{12}$

Q27. MCQ (Single Correct Answer)

Let α, β, γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, (a, b, c $\in$ R and a, b $\neq$ 0). If the system of equations (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$, $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is

Open solution

(A) 5

(B) 3

(C) 1

(D) 0

Q28. MCQ (Single Correct Answer)

The integral $\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{4x^2-4x+6}}dx$ is equal to (where c is a constant of integration)

Open solution

(A) $\frac{1}{2}\sin\sqrt{(2x-1)^2+5}+c$

(B) $\frac{1}{2}\cos\sqrt{(2x+1)^2+5}+c$

(C) $\frac{1}{2}\cos\sqrt{(2x-1)^2+5}+c$

(D) $\frac{1}{2}\sin\sqrt{(2x+1)^2+5}+c$

Q29. MCQ (Single Correct Answer)

The equation of one of the straight lines which passes through the point (1, 3) and makes an angles $\tan^{-1}\left(\sqrt{2}\right)$ with the straight line, $y + 1 = 3\sqrt{2}x$ is :

Open solution

(A) $4\sqrt{2}x + 5y - \left(15 + 4\sqrt{2}\right) = 0$

(B) $5\sqrt{2}x + 4y - \left(15 + 4\sqrt{2}\right) = 0$

(C) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$

(D) $4\sqrt{2}x - 5y - \left(5 + 4\sqrt{2}\right) = 0$

Q30. MCQ (Single Correct Answer)

If $\lim\limits_{x\to 0}\frac{\sin^{-1}x - \tan^{-1}x}{3x^3}$ is equal to L, then the value of (6L + 1) is

Open solution

(A) $\frac{1}{6}$

(B) $\frac{1}{2}$

(C) 6

(D) 2

Q31. MCQ (Single Correct Answer)

A vector $\vec{a}$ has components 3p and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{a}$ has components p + 1 and $\sqrt{10}$, then the value of p is equal to :

Open solution

(A) 1

(B) $-\frac{5}{4}$

(C) $\frac{4}{5}$

(D) -1

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Q32. MCQ (Single Correct Answer)

If the equation $a|z|^2 + \overline{\alpha}z + \alpha\overline{z} + d = 0$ represents a circle where a, d are real constants then which of the following condition is correct?

Open solution

(A) $|\alpha|^2 - ad \neq 0$

(B) $|\alpha|^2 - ad > 0$ and $a \in R - \{0\}$

(C) $|\alpha|^2 - ad \geq 0$ and $a \in R$

(D) $\alpha = 0$, $a, d \in R^+$

Q33. MCQ (Single Correct Answer)

For the four circles M, N, O and P, following four equations are given :

Open solution

Circle M : $x^2 + y^2 = 1$

Circle N : $x^2 + y^2 - 2x = 0$

Circle O : $x^2 + y^2 - 2x - 2y + 1 = 0$

Circle P : $x^2 + y^2 - 2y = 0$

If the centre of circle M is joined with centre of the circle N, further center of circle N is joined with centre of the circle O, centre of circle O is joined with the centre of circle P and lastly, centre of circle P is joined with centre of circle M, then these lines form the sides of a :

(A) Rhombus (B) Square (C) Rectangle (D) Parallelogram

Q34. MCQ (Single Correct Answer)

The real valued function $f(x) = \frac{\cos ec^{-1}x}{\sqrt{x-[x]}}$, where [x] denotes the greatest integer less than or equal to x, is defined for all x belonging to :

Open solution

(A) all real except integers

(B) all non-integers except the interval $[-1, 1]$

(C) all integers except $0, -1, 1$

(D) all real except the interval $[-1, 1]$

Q35. MCQ (Single Correct Answer)

If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, then what is the common domain of the following functions :

Open solution

$f + g$, $f - g$, f/g, g/f, $g - f$ where $(f \pm g)(x) = f(x) \pm g(x), (f/g)x = \frac{f(x)}{g(x)}$

(A) $0 \leq x \leq 1$ (B) $0 \leq x < 1$ (C) $0 < x < 1$ (D) $0 < x \leq 1$

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Q36. MCQ (Single Correct Answer)

If $f(x) = \begin{cases} \frac{1}{|x|} & ; |x| \geq 1 \\ ax^2 + b & ; |x| < 1 \end{cases}$ is differentiable at every point of the domain, then the values of a and b are respectively :

(A) $\frac{1}{2}, \frac{1}{2}$ (B) $\frac{1}{2}, -\frac{3}{2}$ (C) $\frac{5}{2}, -\frac{3}{2}$ (D) $-\frac{1}{2}, \frac{3}{2}$

Open solution

Q37. MCQ (Single Correct Answer)

Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$. If Tr(A) denotes the sum of all diagonal elements of the matrix A, then Tr(A) $-$ Tr(B) has value equal to

(A) 1 (B) 2 (C) 0 (D) 3

Open solution

Q38. MCQ (Single Correct Answer)

The sum of all the 4-digit distinct numbers that can be formed with the digits 1, 2, 2 and 3 is :

(A) 26664 (B) 122664 (C) 122234 (D) 22264

Open solution

Q39. MCQ (Single Correct Answer)

The value of $3 + \cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+....\infty}}}}$ is equal to

(A) 1.5 + $\sqrt{3}$ (B) 2 + $\sqrt{3}$ (C) 3 + 2$\sqrt{3}$ (D) 4 + $\sqrt{3}$

Open solution

Q40. Numerical

The number of times the digit 3 will be written when listing the integers from 1 to 1000 is :

Open solution

Q41. Numerical

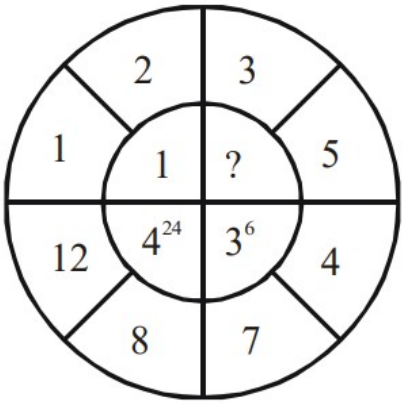
Let f(x) and g(x) be two functions satisfying f(x^2) + g(4 $-$ x) = $4x^3$ and g(4 $-$ x) + g(x) = 0, then the value of $\int_{-4}^{4} f(x^2)dx$ is

Open solution

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Q42. Numerical

The missing value in the following figure is



Open solution

Q43. Numerical

Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral triangle with origin. Then, the value of |a| is :

Open solution

Q44. Numerical

The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is _____.

Open solution

Q45. Numerical

If $f(x) = \int \frac{5x^8+7x^6}{(x^2+1+2x^7)^2}dx, (x \geq 0), f(0) = 0$ and $f(1) = \frac{1}{K}$, then the value of K is

Open solution

Q46. Numerical

A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is _____.

Open solution

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Physics 27 questions

Q47. MCQ (Single Correct Answer)

An oil drop of radius 2 mm with a density 3g cm^{-3} is held stationary under a constant electric field 3.55×10^5 V m^{-1} in the Millikan's oil drop experiment. What is the number of excess electrons that the oil drop will possess? (consider g = 9.81 m/s^2)

Open solution

(A) 48.8×10^{11} (B) 1.73×10^{10} (C) 17.3×10^{10} (D) 1.73×10^{12}

Q48. MCQ (Single Correct Answer)

Imagine that the electron in a hydrogen atom is replaced by a muon (μ). The mass of muon particle is 207 times that of an electron and charge is equal to the charge of an electron. The ionization potential of this hydrogen atom will be :

Open solution

(A) 13.6 eV (B) 2815.2 eV (C) 331.2 eV (D) 27.2 eV

Q49. MCQ (Single Correct Answer)

A plane electromagnetic wave of frequency 100 MHz is travelling in vacuum along the x-direction. At a particular point in space and time, $\overrightarrow{B} = 2.0 \times 10^{-8}\hat{k}T$. (where, $\hat{k}$ is unit vector along z-direction) What is $\overrightarrow{E}$ at this point?

Open solution

(A) 0.6 $\hat{j}$ V/m (B) 6.0 $\hat{k}$ V/m (C) 6.0 $\hat{j}$ V/m (D) 0.6 $\hat{k}$ V/m

Q50. MCQ (Single Correct Answer)

A thin circular ring of mass M and radius r is rotating about its axis with an angular speed ω. Two particles having mass m each are now attached at diametrically opposite points. The angular speed of the ring will become :

Open solution

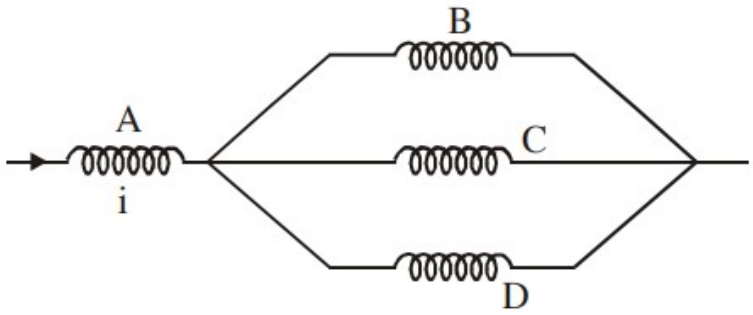
(A) $\omega\frac{M}{M+m}$ (B) $\omega\frac{M+2m}{M}$ (C) $\omega\frac{M}{M+2m}$ (D) $\omega\frac{M-2m}{M+2m}$

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Q51. MCQ (Single Correct Answer)

Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is 3T, the field at the center of C would be : (Assume that the magnetic field is confined with in the volume of respective solenoid).

Open solution

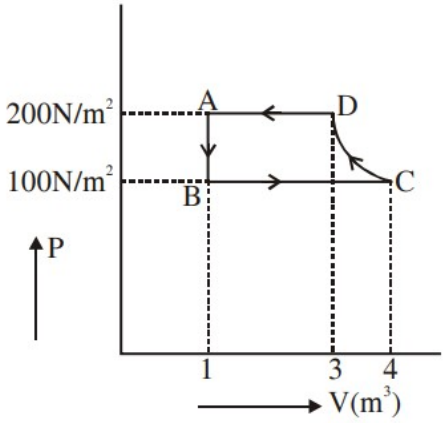


(A) 12T (B) 6T (C) 9T (D) 1T

Q52. MCQ (Single Correct Answer)

The P-V diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use γ = 1.4) :

Open solution



(A) -500 J (B) -400 J (C) 400 J (D) 200 J

Q53. MCQ (Single Correct Answer)

In Young's double slit arrangement, slits are separated by a gap of 0.5 mm, and the screen is placed at a distance of 0.5 m from them. The distance between the first and the third bright fringe formed when the slits are illuminated by a monochromatic light of 5890 $\overset{o}{A}$ is :-

Open solution

(A) 1178×10^{-9} m (B) 1178×10^{-6} m (C) 1178×10^{-12} m (D) 5890×10^{-7} m

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Q54. MCQ (Single Correct Answer)

Open solution

A particle is travelling 4 time as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2 : 1, the mass of the particle is :

(A) $\frac{1}{16}$ times the mass of e^-

(B) 8 times the mass of e^-

(C) 16 times the mass of e^-

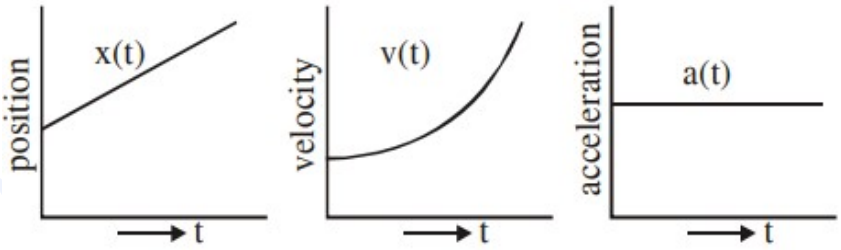
(D) $\frac{1}{8}$ times the mass of e^-

Q55. MCQ (Single Correct Answer)

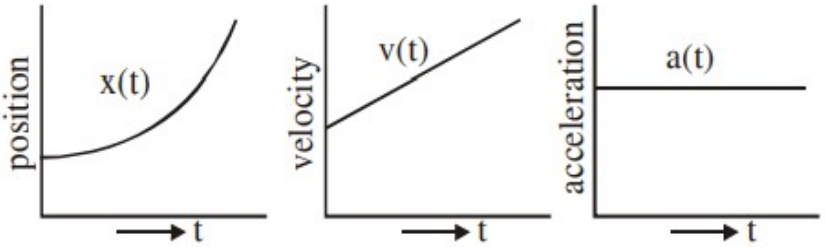
Open solution

The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by :

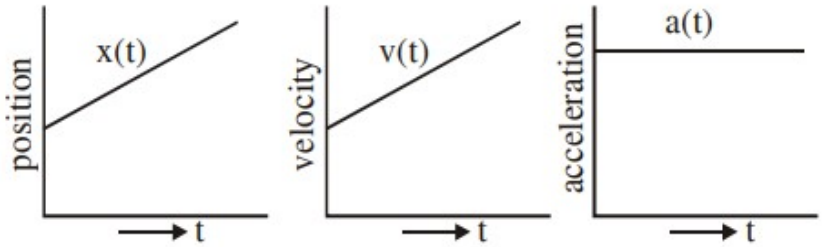
(A)



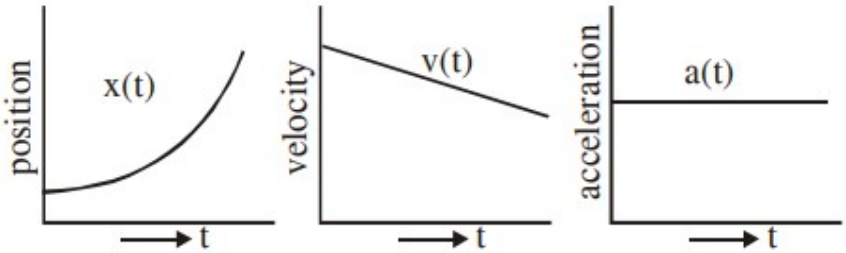
(B)



(C)



(D)



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Q56. MCQ (Single Correct Answer)

In the experiment of Ohm's law, a potential difference of 5.0 V is applied across the end of a conductor of length 10.0 cm and diameter of 5.00 mm. The measured current in the conductor is 2.00 A. The maximum permissible percentage error in the resistivity of the conductor is :

(A) 3.9 (B) 8.4 (C) 7.5 (D) 3.0

Open solution

Q57. MCQ (Single Correct Answer)

In a series LCR resonance circuit, if we change the resistance only, from a lower to higher value :

(A) The bandwidth of resonance circuit will increase.

(B) The resonance frequency will increase.

(C) The quality factor will increase.

(D) The quality factor and the resonance frequency will remain constant.

Open solution

Q58. MCQ (Single Correct Answer)

An AC source rated 220 V, 50 Hz is connected to a resistor. The time taken by the current to change from its maximum to the rms value is :

(A) 2.5 ms (B) 25 ms (C) 2.5 s (D) 0.25 ms

Open solution

Q59. MCQ (Single Correct Answer)

Your friend is having eye sight problem. She is not able to see clearly a distant uniform window mesh and it appears to her as non-uniform and distorted. The doctor diagnosed the problem as :

(A) Astigmatism (B) Myopia with Astigmatism

(C) Presbyopia with Astigmatism (D) Myopia and Hypermetropia

Open solution

Q60. MCQ (Single Correct Answer)

A loop of flexible wire of irregular shape carrying current is placed in an external magnetic field. Identify the effect of the field on the wire.

(A) Loop assumes circular shape with its plane normal to the field.

(B) Loop assumes circular shape with its plane parallel to the field.

(C) Wire gets stretched to become straight.

(D) Shape of the loop remains unchanged.

Open solution

Q61. MCQ (Single Correct Answer)

The time period of a satellite in a circular orbit of radius R is T. The period of another satellite in a circular orbit of radius 9R is :

(A) 9 T (B) 27 T (C) 12 T (D) 3 T

Open solution

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Q62. Numerical

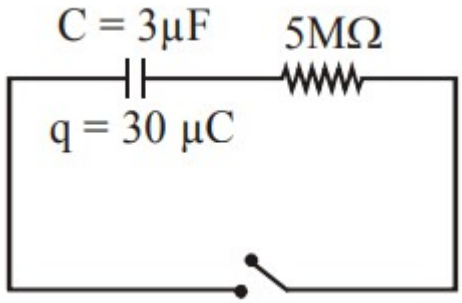
A particle performs simple harmonic motion with a period of 2 second. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}$s. The value of 'a' to the nearest integer is _____.

Open solution

Q63. Numerical

The circuit shown in the figure consists of a charged capacitor of capacity 3 μF and a charge of 30 μC. At time t = 0, when the key is closed, the value of current flowing through the 5 MΩ resistor is 'x' μ-A. The value of 'x' to the nearest integer is _____.

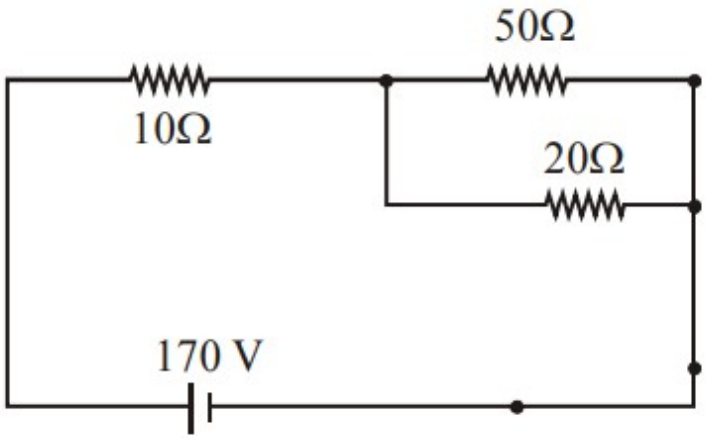
Open solution



Q64. Numerical

The voltage across the 10Ω resistor in the given circuit is x volt.

Open solution



The value of 'x' to the nearest integer is _____.

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Q65. Numerical

Two separate wires A and B are stretched by 2 mm and 4 mm respectively, when they are subjected to a force of 2 N. Assume that both the wires are made up of same material and the radius of wire B is 4 times that of the radius of wire A. The length of the wires A and B are in the ratio of a : b. Then a/b can be expressed as 1/x where x is _____.

Open solution

Q66. Numerical

A person is swimming with a speed of 10 m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is 'x' m/s. The value of 'x' to the nearest integer is _____.

Open solution

Q67. Numerical

A parallel plate capacitor has plate area 100 m^2 and plate separation of 10 m. The space between the plates is filled up to a thickness 5 m with a material of dielectric constant of 10. The resultant capacitance of the system is 'x' pF.

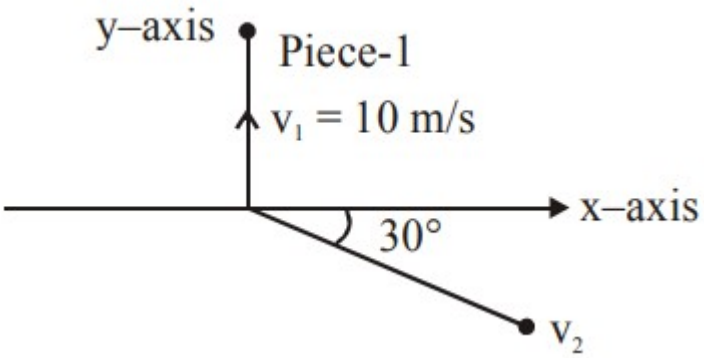
The value of $\varepsilon_0 = 8.85 \times 10^{-12}$ F.m^{-1}.

The value of 'x' to the nearest integer is _____.

Open solution

Q68. Numerical

A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m/s along the x-axis, hits another ball of mass 20 kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along y-axis with a speed of 10 m/s. The second piece starts moving at an angle of 30° with respect to the x-axis. The velocity of the ball moving at 30° with x-axis is x m/s. The configuration of pieces after collision is shown in the figure below. The value of x to the nearest integer is _____.



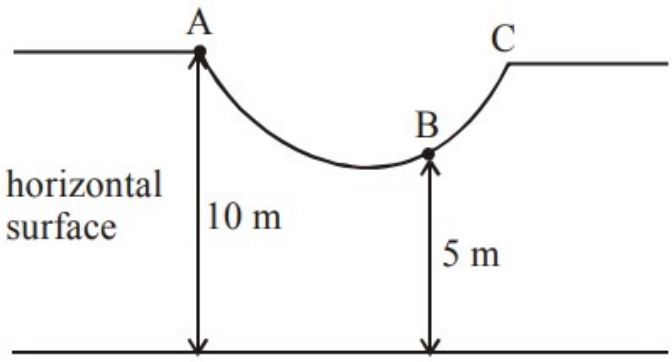
Open solution

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Q69. Numerical

As shown in the figure, a particle of mass 10 kg is placed at a point A. When the particle is slightly displaced to its right, it starts moving and reaches the point B. The speed of the particle at B is x m/s. (Take g = 10 m/s^2)

The value of 'x' to the nearest integer is _____.



Open solution

Q70. Numerical

A bullet of mass 0.1 kg is fired on a wooden block to pierce through it, but it stops after moving a distance of 50 cm into it. If the velocity of bullet before hitting the wood is 10 m/s and it slows down with uniform deceleration, then the magnitude of effective retarding force on the bullet is 'x' N. The value of 'x' to the nearest integer is _____.

Open solution

Q71. MCQ (Single Correct Answer)

The time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$. The measured value of the length of pendulum is 10 cm known to a 1mm accuracy. The time for 200 oscillations of the pendulum is found to be 100 second using a clock of 1s resolution. The percentage accuracy in the determination of 'g' using this pendulum is 'x'. The value of 'x' to be nearest integer is :-

(A) 2% (B) 3% (C) 5% (D) 4%

Open solution

Q72. MCQ (Single Correct Answer)

A constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the box in time 't' is proportional to :-

(A) $t^{2/3}$ (B) $t^{3/2}$ (C) t (D) $t^{1/2}$

Open solution

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Q73. MCQ (Single Correct Answer)

What will be the average value of energy along one degree of freedom for an ideal gas in thermal equilibrium at a temperature T? (k_B is Boltzmann constant)

(A) $\frac{1}{2}k_BT$ (B) $\frac{2}{3}k_BT$ (C) $\frac{3}{2}k_BT$ (D) k_BT

Open solution

Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match the list - I with list - II

	List - I (Class of Drug)		**List - II (Example)**
(a)	Antacid	(i)	Novestrol
(b)	Artificial sweetener	(ii)	Cimetidine
(c)	Antifertility	(iii)	Valium
(d)	Tranquilizers	(iv)	Alitame

Open solution

(A) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

(B) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

Q75. MCQ (Single Correct Answer) · **Chemistry**

The statements that are TRUE :

(A) Methane leads to both global warming and photochemical smog

(B) Methane is generated from paddy fields

(C) Methane is a stronger global warming gas than CO_2

(D) Methane is a part of reducing smog

Choose the most appropriate answer from the options given below :

Open solution

(A) (A), (B), (C) only **(B)** (A) and (B) only

(C) (B), (C), (D) only **(D)** (A), (B), (D) only

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II

	List - I		List - II
(a)	$Ca(OCl)_2$	(i)	Antacid
(b)	$CaSO_4.\frac{1}{2}H_2O$	(ii)	Cement
(c)	CaO	(iii)	Bleach
(d)	$CaCO_3$	(iv)	Plaster of paris

Choose the most appropriate answer from the options given below :

(A) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

(B) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

(C) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(D) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The number of ionisable hydrogens present in the product obtained from a reaction of phosphorus trichloride and phosphonic acid is :

(A) 3 **(B)** 0 **(C)** 2 **(D)** 1

Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

In a binary compound, atoms of element A form a hcp structure and those of element M occupy 2/3 of the tetrahedral voids of the hcp structure. The formula of the binary compound is :

(A) M_2A_3 **(B)** M_4A_3 **(C)** M_4A **(D)** MA_3

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The chemical that is added to reduce the melting point of the reaction mixture during the extraction of aluminium is :

(A) Cryolite **(B)** Bauxite **(C)** Calamine **(D)** Kaolite

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The differential equation satisfied by the system of parabolas

$y^2 = 4a(x + a)$ is :

(A) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) - y = 0$

(B) $y\left(\frac{dy}{dx}\right)^2 - 2x\left(\frac{dy}{dx}\right) + y = 0$

(C) $y\left(\frac{dy}{dx}\right)^2 + 2x\left(\frac{dy}{dx}\right) - y = 0$

(D) $y\left(\frac{dy}{dx}\right) + 2x\left(\frac{dy}{dx}\right) - y = 0$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

Let $(1 + x + 2x^2)^{20} = a_0 + a_1x + a_2x^2 + ... + a_{40}x^{40}$. Then $a_1 + a_3 + a_5 + ... + a_{37}$ is equal to

(A) $2^{20}(2^{20} - 21)$ (B) $2^{19}(2^{20} - 21)$ (C) $2^{19}(2^{20} + 21)$ (D) $2^{20}(2^{20} + 21)$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Choose the correct statement about two circles whose equations are given below :

$x^2 + y^2 - 10x - 10y + 41 = 0$

$x^2 + y^2 - 22x - 10y + 137 = 0$

(A) circles have same centre

(B) circles have no meeting point

(C) circles have only one meeting point

(D) circles have two meeting points

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If α, β are natural numbers such that

$100^{\alpha} - 199\beta = (100)(100) + (99)(101) + (98)(102) + ... + (1)(199)$, then the slope of the line passing through (α, β) and origin is :

(A) 540 (B) 550 (C) 530 (D) 510

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

$\frac{1}{3^2-1} + \frac{1}{5^2-1} + \frac{1}{7^2-1} + ... + \frac{1}{(201)^2-1}$ is equal to

(A) $\frac{101}{404}$ (B) $\frac{25}{101}$ (C) $\frac{101}{408}$ (D) $\frac{99}{400}$

Open solution

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Q85. Numerical · **Mathematics**

Let the plane ax + by + cz + d = 0 bisect the line joining the points (4, −3, 1) and (2, 3, −5) at the right angles. If a, b, c, d are integers, then the minimum value of $(a^2 + b^2 + c^2 + d^2)$ is _____.

Open solution

Q86. Numerical · **Mathematics**

The equation of the planes parallel to the plane x − 2y + 2z − 3 = 0 which are at unit distance from the point (1, 2, 3) is ax + by + cz + d = 0. If (b − d) = k(c − a), then the positive value of k is :

Open solution

Q87. Numerical · **Mathematics**

The number of solutions of the equation

$|\cot x| = \cot x + \frac{1}{\sin x}$ in the interval [0, 2π] is

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

Match List - I with List - II.

List - I

(a) 10 km height over earth's surface

(b) 70 km height over earth's surface

(c) 180 km height over earth's surface

(d) 270 km height over earth's surface

List - II

(i) Thermosphere

(ii) Mesosphere

(iii) Stratosphere

(iv) Troposphere

(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(C) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

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Q89. Numerical · **Physics**

An npn transistor operates as a common emitter amplifier with a power gain of 10^6. The input circuit resistance is 100Ω and the output load resistance is 10 KΩ. The common emitter current gain 'β' will be _____. (Round off to the Nearest Integer).

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

A radioactive sample disintegrates via two independent decay processes having half lives $T_{1/2}^{(1)}$ and $T_{1/2}^{(2)}$ respectively. The effective half-life $T_{1/2}$ of the nuclei is :

Open solution

(A) None of the above

(B) $T_{1/2} = T_{1/2}^{(1)} + T_{1/2}^{(2)}$

(C) $T_{1/2} = \frac{T_{1/2}^{(1)} T_{1/2}^{(2)}}{T_{1/2}^{(1)} + T_{1/2}^{(2)}}$

(D) $T_{1/2} = \frac{T_{1/2}^{(1)} + T_{1/2}^{(2)}}{T_{1/2}^{(1)} - T_{1/2}^{(2)}}$

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Answer Key

Chemistry

1. **B**	2. **B**	3. **C**	4. **D**	5. **D**
6. **C**	7. **B**	8. **B**	9. **D**	10. **D**
11. **B**	12. **C**	13. **A**	14. **D**	15. **15**
16. **10**	17. **3**	18. **128**	19. **16**	20. **3**
21. **3**	22. **50**	23. **46**	24. **18**	

Mathematics

25. **B**	26. **D**	27. **B**	28. **A**	29. **A**
30. **D**	31. **D**	32. **B**	33. **B**	34. **B**
35. **C**	36. **D**	37. **B**	38. **A**	39. **A**
40. **300**	41. **512**	42. **4**	43. **6**	44. **35**
45. **4**	46. **80**			

Physics

47. **B**	48. **B**	49. **C**	50. **C**	51. **D**
52. **A**	53. **B**	54. **D**	55. **B**	56. **A**
57. **A**	58. **A**	59. **B**	60. **A**	61. **B**
62. **6**	63. **2**	64. **70**	65. **32**	66. **5**
67. **161**	68. **20**	69. **10**	70. **10**	71. **B**
72. **B**	73. **A**			

Out of syllabus

74. **A**	75. **A**	76. **C**	77. **C**	78. **B**
79. **A**	80. **C**	81. **B**	82. **C**	83. **B**
84. **B**	85. **28**	86. **4**	87. **1**	88. **A**
89. **100**	90. **C**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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